ZENTRAL HACK 2026 · AGENT HARNESS

Spec in.
Strategy out.



An agent harness for games it has never seen.

Elias · Isaak · Timo

THE TASK · THE INSIGHT

An unseen game, a fixed budget, only wins count

The task

A $0.03 · 3 min

B $0.50 · 10 min

C $10 · 30 min

What research says

9 / 10

games where code written by an LLM matches or beats an LLM picking the moves

Code World Models, Google DeepMind

Our insight

The LLM is the programmer, not the player.

Everything it writes gets tested by actually running it.

Sources: Zentral Hack 2026 task description; Code World Models for General Game Playing, arXiv 2510.04542.

THE HARNESS

The LLM writes, our code verifies

spec.md in

strategy.py out

1
Write
drafts + a simulator

2
Verify
against the spec

3
Self-play
real tournament rules

4
Improve
challengers vs. champion

5
Guard
never a forfeit

↻ repeat while time and money last

HOW WE BUILT IT

Built like it works: measure, decide, repeat

10

games, 4 never tuned on

120+

measured runs

≈ $45

API cost

Built with Claude Code · 416 automated tests

EPOCH 1

Ablation

kept only what wins

EPOCH 2

Tournament conditions

selection = reality

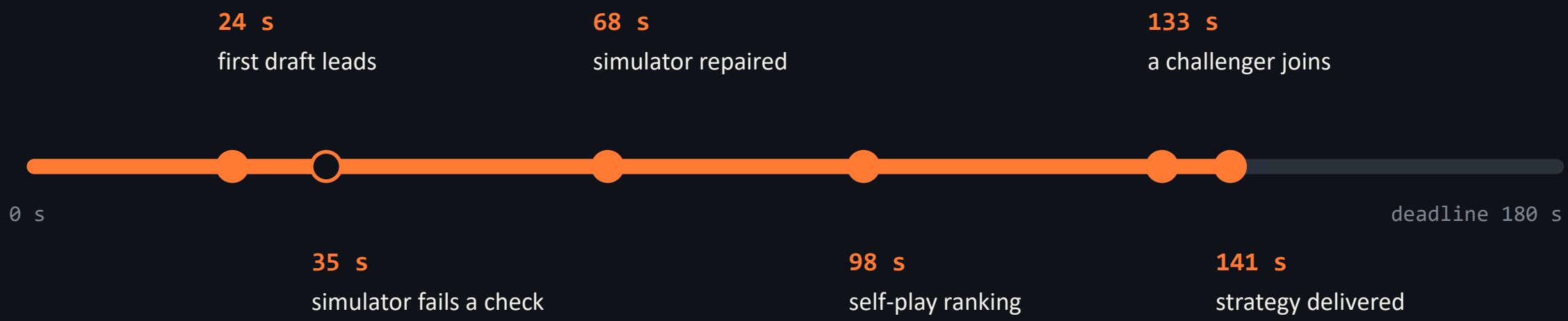
EPOCH 3

Midpoint feedback ·2 experiments

2 ideas rejected, weak runs traced and fixed

ONE REAL RUN, FROM SPEC TO STRATEGY.PY

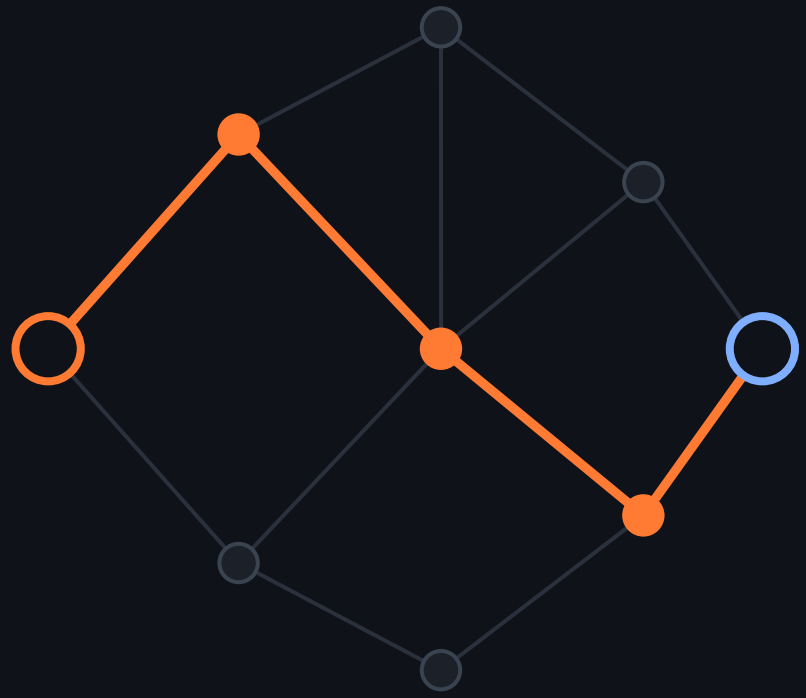
Spec in. Strategy out. Never a forfeit.



141 seconds, 1.8 cents, one champion. A real task-A run in the evaluation's container: drafts written, the simulator checked against the spec and repaired, drafts ranked by self-play, a challenger tried, the champion delivered well before the deadline.

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Thank you for listening!



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